

September, 2026

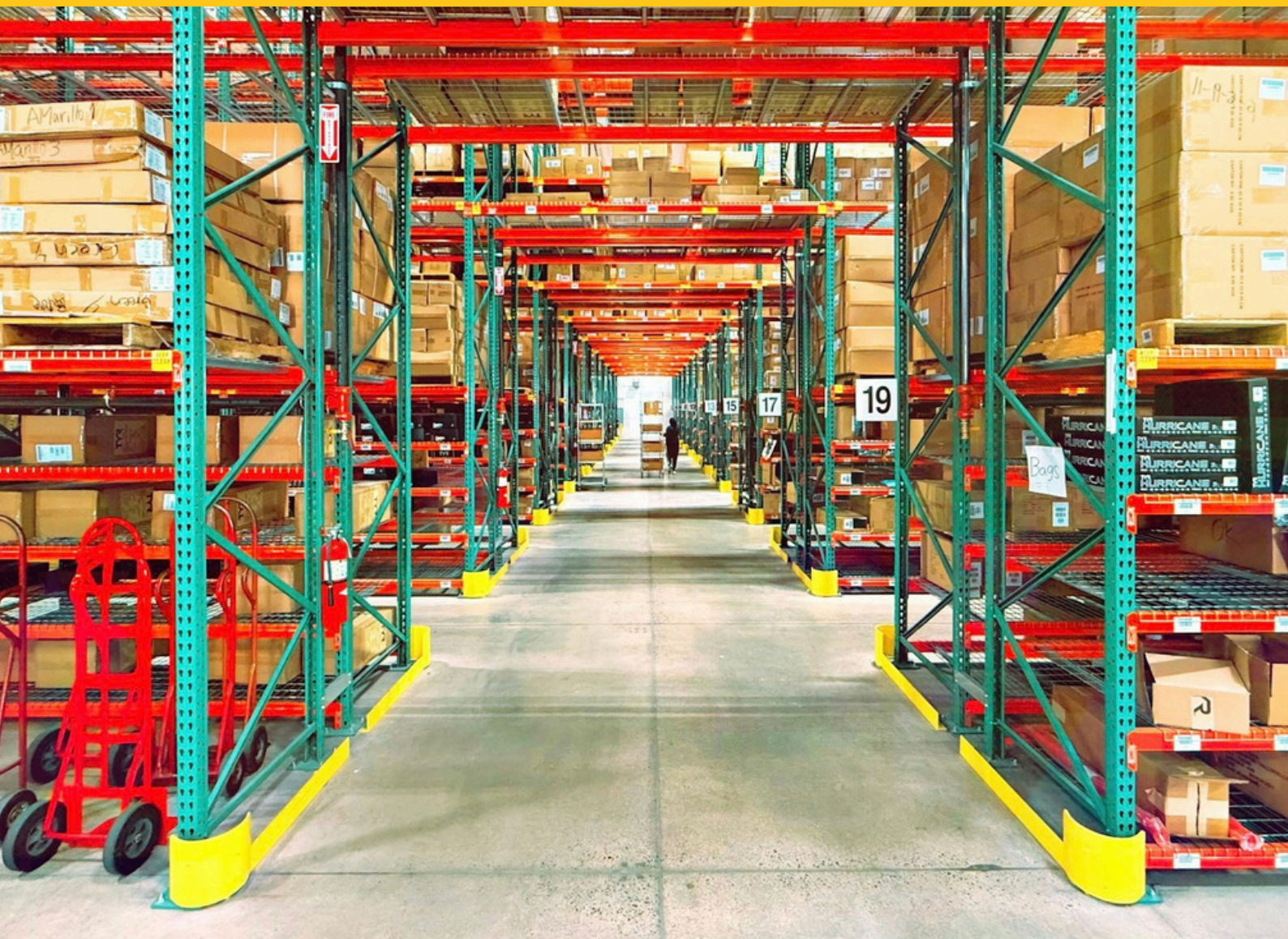
Everything & Then Some: Predicting Repeat Purchases with ML

Group Project

Machine Learning 2026/2027

Due: December 22nd, 2026

(17:59)



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I. Introduction

Everything & Then Some, Ltd. is a diversified wholesale company supplying businesses and specialist retailers across a broad portfolio of products, including tools and hardware, lighting, automotive care, household goods and other specialist product lines. Its customers show very different purchasing patterns, making repeat-purchase behaviour difficult to anticipate.

The company wants to use its historical sales data to support more targeted commercial actions. Your team has been asked to **develop a predictive model capable of answering the following question: will this customer make another purchase within the next 90 days after the current transaction?**

II. Project Goals

The goal of your project is four-fold:

1. **Preprocessing and Exploratory Data Analysis (EDA):** **clean** and **explore** the supplied transaction-level **data**, identify inconsistencies and prepare it for modelling.
2. **Binary Classification & Optimization:** **predict whether a customer will make another purchase within 90 days** of the current purchase. **Define a consistent model-assessment strategy**, compare candidate models, perform feature selection and optimize the strongest approaches.
3. **Kaggle Competition:** **generate predictions for every observation** in test.csv and **submit them using the provided sample_submission.csv structure**.
4. **Additional Insights:** this section is open-ended. Possible directions include **feature importance and prediction errors**, **ablation study**, **prediction probabilities and calibration**, **alternative feature sets**, or a **simple analytics interface for new observations**

02 III. Dataset

You have access to three data files and one sample submission file:

train.csv contains 6,534 historical purchase occasions and the binary target repeat_purchase_90d. test.csv contains 986 later purchase occasions with the same predictors but without the target. clientes.csv is an optional customer-level table that can be joined through customer_id. sample_submission.csv provides the required submission structure and a simple baseline prediction.

Each row in train/test represents a customer purchase occasion. The target is 1 when the same customer makes another purchase within the following 90 days and 0 otherwise. The available data contains the following attributes and attribute groups:

Attribute	Description
ID	Unique modelling-row and Kaggle submission identifier.
customer_id	Anonymized customer identifier; join key for clientes.csv.
repeat_purchase_90d	Target: 1 if the customer purchases again within 90 days; 0 otherwise. Train only.
purchase_date	Anonymized purchase date. Temporal order and intervals are preserved.
Current monetary values	Current transaction totals, tax, shipping, discounts and cost.
Current purchase detail	Current detail count and number of invoices in the purchase occasion.
Current categoricals	Payment, salesperson, zone, transport, warehouse and document series.
Historical timing	Recency, tenure, prior purchase count and historical purchase gaps.
Historical value	Prior spend and prior purchase-value summary statistics.
Rolling frequency	Purchase counts over the previous 30, 90, 180 and 365 days.
Rolling spend	Customer spend over the previous 30, 90, 180 and 365 days.
random_noise	Additional numerical field; assess whether it contains useful information.
clientes.csv	Optional customer-level attributes joined through customer_id.

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IV. Outline

Your project deliverables (notebook or zip of notebooks) should respect the following outline:

Group Member Contribution

What part(s) of the work were done by each member and an estimated % contribution of each member towards the final work.

Abstract

A small summary of your work (200 to 300 words). The abstract should give an overview of your work: What is the context? What are your goals? What did you do? What were your main results, and what conclusions did you draw from them?

I. Identifying Business Needs

- Overview and main goals of the project
- Description of the overall process and identification of model assessment approach adopted in the work (CV, LOO, Holdout, etc...)

II. Data Exploration and Preprocessing

- Description of data received -> key insights
- Steps taken to clean and prepare the data based on exploration

III. Binary Classification & Optimization

- Explanation of model assessment strategy and metrics used
- Feature Selection Strategy and results
- Optimization efforts: presentation, results and discussion
- Comparison of performance between candidate models

IV. Open-Ended Section

- Objectives for the Section and description of the actions taken
- Results and discussion of main findings → key takeaways

Note: This section expects that the objectives set go beyond what would reasonably be considered as adding or removing techniques to your pipeline. (e.g., using a feature selection technique not covered in class on your regular pipeline is not sufficient, but explicitly comparing different feature sets would be).

V. Deployment

The final section of your work should implement the pipeline to generate reliable predictions for new data. The output should be the .csv file that you consider the solution you selected on Kaggle as your best.

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V. Deliverables

Upon the project's deadline, you will be required to submit:

- A clean Jupyter notebook (or a zipfile) featuring all the code you used throughout the project to:
 - a. Decide on your final solution
 - b. Obtain your final results (code that helped you make decisions, but does not directly contribute to reaching the goal, should be included, but commented).

VI. Evaluation

Your work will be evaluated according to the following criteria:

Criteria	Percentage (%)	Maximum Grade (out of 20)
Project Structure and Notebook(s) Quality	20	4
Data Exploration & Initial Preprocessing	20	4
Classification Benchmarking and Optimization	35	7
Open-Ended Section	20	4
Deployment	5	1
Extra Point: Have Project Be Publicly Available on GitHub	-	1

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Your grade will reflect our assessment of the quality of your work in terms of quality of writing, clarity, conciseness, correctness and efficiency. Please find below more details about what is taken into account for each topic:

- Project Structure and Notebook(s) Quality (4v): Your notebook(s) should be readable and understandable to someone reading and looking at them for the first time. Every cell you present should have a clear purpose. This section also encompasses the overall quality of your introduction and conclusions for each cell.
- Data Exploration & Initial Preprocessing (4v): Describe the data and extract meaningful insights that you consider helpful. Avoid adding visualizations and elements that do not address the problem at hand. Additionally, it should clearly explain the steps and rationale behind the data cleaning process to make the data usable by your predictive models.
- Classification Benchmarking and Optimization (7v): Describe your strategy for model assessment. This section is separated into different components:
 - Kaggle Performance: 1v
 - Additional Preprocessing: 0.5v
 - Feature Selection Strategy and Implementation: 1.5v
 - Modelling approach - model assessment strategy (holdout, cross-validation, etc...) and algorithms (minimum of 5 covered in class) used: 2v
 - Performance assessment - rationale for choice of evaluation metric(s) and interpretation of results: 1.5v
 - Model optimization: 1v
- Open-Ended Section (4v): Describe your strategy for the additional insights objective. This section is separated into different components:
 - Formulation and Adequacy of the Objectives: 0.5v
 - Difficulty of tasks: 1v
 - Correctness/efficiency of implementation: 1v
 - Discussion of results: 1v
 - Alignment between results and communicated objectives: 0.5v
- Deployment (1v): In the final part of the project, you should be able to take your test data and ensure that it follows the steps implemented in the previous stages of the project.

06 VII. Parting Notes

1. Deliveries made after the deadline will incur a penalty of 1 point per day. Deliveries made before the deadline will receive a bonus of 0.15 points per day of delivery in advance (up to a maximum of 1 point).
2. If you use automated packages (e.g., ydata, pandas-profiling), we will only consider observations that have a visible impact on the project (i.e., a key insight that is acted upon).
3. When it comes to creating Predictive Models, you are allowed to create your own implementations or use one of the following packages: vanilla Scikit-Learn, Tensorflow, Keras, PyTorch (and skorch) or Transformers.
4. Using Lazy Predict or similar AutoML (e.g., Feature Tools, TSMFRESH) packages is explicitly forbidden.
5. We won't consider any steps or results that are not mentioned in any of your submitted notebooks.
6. Everything in your clean notebook must have a clear purpose. Avoid including irrelevant, unimportant, or redundant information. Additionally, please refrain from providing theoretical explanations of topics covered in class.
7. The trustworthiness of the information you provide is key. You should look to source information from peer-reviewed journals (thus, avoid citing Medium, TowardsDataScience and similar sources). Also, for the love of God, DO NOT CITE ChatGPT or any other LLM for that matter.
8. Before submitting, run your notebook from the start one last time (if you used a GridSearch, you can comment this cell, but you should run the final model with the GS parameters in a different cell).
9. If you tested more elements that should be considered, but did not feature in your clean notebook, please submit the code you used in a separate submission form created for that purpose.
10. We will run your Jupyter Notebooks. Please ensure that we can run the notebook from start to finish without interruption. Notebooks that do not fulfil this condition will be severely penalized.
11. The code will pass through a process of plagiarism and AI generation checking.
12. You must submit to the Kaggle competition to get points for that component.
13. When determining the grade for your work, there will be a comparative component between it and the work presented by your peers.

Friendly Reminders:

1. The order in which the group members present their notebooks is decided by the professors at the start of the discussion. Not attending will result in a 0.
2. If something is good enough to be mentioned, it is also good enough to know. DO NOT include techniques/algorithms/steps you cannot explain in your work: we may (and probably will) ask about them in the discussion.
3. Finished is better than perfect.